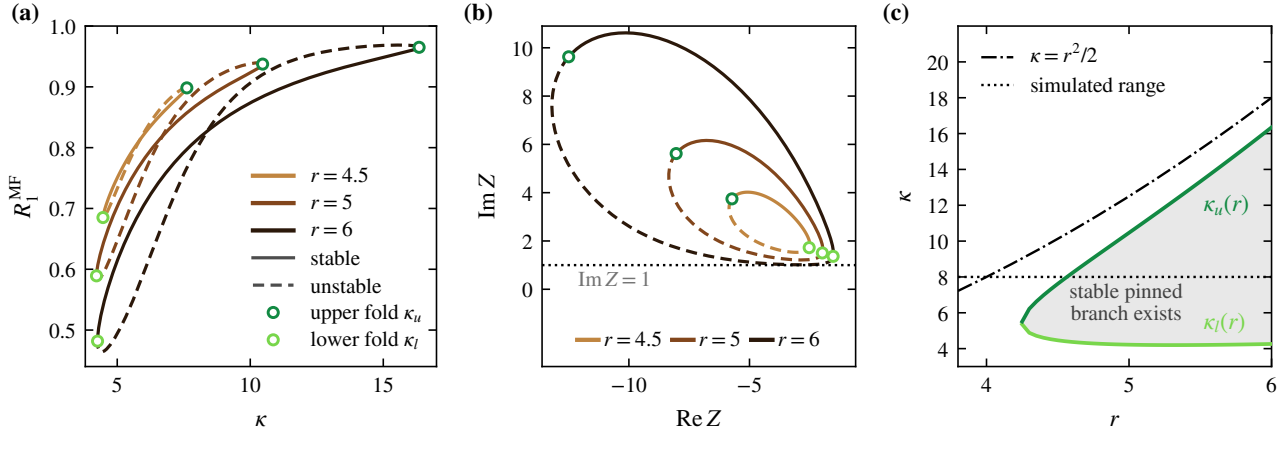
(a)
1.0
0.9
0.8
0.7
0.6
0.5
R_1^{MF}
5
10
15
κ
$r = 4.5$
$r = 5$
$r = 6$
stable
unstable
upper fold κ_u
lower fold κ_l
(b)
10
8
6
4
2
0
Im Z
$\mathrm{Im}\, Z = 1$
$r = 4.5$ $r = 5$ $r = 6$
−10
−5
Re Z
(c)
$\kappa = r^2/2$
simulated range
20
18
16
14
12
10
8
6
4
κ
$\kappa_u(r)$
stable pinned
branch exists
$\kappa_l(r)$
4
5
6
r